Use together with the General sheet (Sheet 1).

Record measured outcomes and empirical findings reported in RCTs, pilots, field studies, implementation studies, and other real-world evaluations identified in Sheet 1.

Variables to extract

1 Study / Source

Record the study or source identifier used in Sheet 1. Distinguish separate deployments when one source reports more than one intervention.

2 AI Tool / Program

Record the AI tool or programme name consistently with Sheet 1.

3 Outcome Domain

Definition: Standardized category describing the type of outcome measured. Use one of the following categories whenever possible.

Adoption

Whether eligible farmers, agricultural workers, farms, or organizations begin using the AI tool or AI-enabled service.

Examples:

- share of eligible farmers enrolling in an AI advisory service
- share of extension workers who activated a chatbot
- share of farms starting to use an AI irrigation system
- share of producers using AI-supported disease identification at least once

Keywords / cues: adopted, take-up, consented, enrolled, activated, onboarded, started using, participation rate

Usage

How much, how often, or how long the AI tool is actually used after access/adoption.

Examples:

- crop images analysed per week
- queries per farmer or extension worker
- active users, sessions, continued use, or retention
- number of irrigation decisions supported by AI

Keywords / cues: usage, frequency, volume, queries, messages, active users, logins, retention

Diagnostic / Predictive Accuracy

Agreement between AI outputs and a reference standard in a real agricultural setting, or correctness of AI-generated agricultural information.

Examples:

- sensitivity and specificity for a crop disease
- precision, recall, F1, or area under the ROC curve
- yield-forecast RMSE or MAE against observed harvest records
- agreement with agronomists or laboratory testing
- share of chatbot answers judged correct by agricultural experts

Keywords / cues: sensitivity, specificity, accuracy, agreement, precision, recall, prediction error, RMSE, MAE

Technical Performance / Data Quality

Whether the AI system functioned as intended in routine agricultural conditions.

Examples:

- share of crop images rejected as unusable
- sensor failure or system downtime
- processing time or response latency
- share of questions the system could not answer

Keywords / cues: image quality, rejected, failure rate, downtime, latency, unanswered, missing sensor data

Problem Detection / Service Yield

Changes in the number or proportion of agricultural problems identified, assessments completed, or service actions generated by the diagnostic or advisory process.

Examples:

- pest infestations or disease cases detected
- farms assessed or field inspections triggered
- veterinary or agronomic referrals generated
- unnecessary confirmatory inspections avoided

Keywords / cues: detection, cases identified, service yield, inspections, referrals, assessments completed

“Service yield” refers to the output of the service, not crop yield.

Action Continuum / Linkage

Whether users move from an AI output to the next intended agricultural-service or management step.

Examples:

- completion of a recommended field inspection or specialist referral
- time from pest alert to field confirmation or treatment
- uptake of an input or extension service following a recommendation
- follow-up completion after an AI-generated alert

Keywords / cues: linkage, referral uptake, time to action, response, completion, follow-up

Farmer Behaviour / Programme Engagement

Changes in farming-related behaviour, knowledge, or engagement with an agricultural programme among farmers or beneficiaries.

Examples:

- listening to agricultural voice messages
- drop-off from an advisory programme
- farmer knowledge scores
- adoption of recommended planting, irrigation, or pest-management practices

Keywords / cues: engagement, knowledge, behaviour, drop-off, practice adoption, adherence to advice

Agricultural Production Outcome

Changes in actual agricultural output, crop or animal condition, or product quality.

Examples:

- harvested crop yield or total production
- crop loss or field disease incidence
- milk production, animal growth, or aquaculture survival
- product quality, marketable output, or post-harvest loss

Keywords / cues: yield, harvest, production, crop loss, growth, mortality, survival, product quality

Resource Use / Environmental Outcome

Changes in agricultural input use, efficiency, or measured environmental conditions.

Examples:

- irrigation water, fertilizer, pesticide, feed, or energy use
- water-use or nutrient-use efficiency
- soil indicators, runoff, or emissions
- measured biodiversity outcomes

Keywords / cues: water saved, input use, fertilizer, pesticide, efficiency, soil, emissions, runoff, biodiversity

Livelihood / Food Security Outcome

Changes in producer or household economic welfare, food security, or livelihood resilience.

Examples:

- farm revenue, gross margin, net profit, or household income
- household food-security score or dietary diversity

- variability of income or losses during a weather shock

Keywords / cues: income, profit, revenue, livelihoods, food security, dietary diversity, resilience

Agricultural-Worker Practice / Knowledge

Changes in extension-worker, agronomist, veterinarian, or operator decisions, advice, practices, or knowledge.

Examples:

- adherence to agronomic guidance
- changes in treatment or referral decisions
- agricultural-worker knowledge score
- quality of extension advice

Keywords / cues: guideline adherence, practice, decision change, knowledge, advice quality

Agricultural-Worker Productivity / Workload

Changes in human time, workload, output, or efficiency.

Examples:

- farms served per extension worker
- time required to assess a crop image
- labor hours saved per hectare
- consultations completed per worker per day

Keywords / cues: productivity, workload, time saved, throughput, labor, farms per worker

Service Efficiency / Timeliness

Changes in the speed or efficiency of the service from the farmer's or service provider's perspective.

Examples:

- waiting time for advice
- time to diagnostic result
- turnaround time for an agricultural-service request
- same-day advice or input delivery

Keywords / cues: waiting time, turnaround, duration, time to result, same-day

Farmer / User Experience

Satisfaction, trust, or acceptability reported by people who actually used the AI-enabled service.

Examples:

- farmer satisfaction score
- trust in AI advice

- perceived usability among participating extension workers

Keywords / cues: satisfaction, experience, trust, acceptability among users, usability

Access

Whether the intervention increases access to agricultural advice, diagnostics, inputs, markets, finance, or other agricultural services.

Examples:

- farmers reached in areas without extension staff
- coverage of the eligible population
- availability of advice in local languages or remote areas

Keywords / cues: access, reach, coverage, availability, underserved, rural

Equity

Differences in adoption, usage, accuracy, or outcomes across groups.

Examples:

- women vs men
- smaller vs larger farms
- tenants vs landowners
- remote vs better-connected farmers
- low-literacy vs higher-literacy users

Keywords / cues: heterogeneous effects, subgroup, disadvantaged, equity, disparity, performed worse among

Safety / Harms

Adverse or unintended consequences of using the AI.

Examples:

- missed pest or disease cases and their consequences
- unnecessary input use following false-positive alerts
- inappropriate or unsafe agronomic advice
- crop damage, animal harm, worker exposure, or financial loss
- privacy or farm-data incidents

Keywords / cues: safety, harm, adverse event, false negative, missed, unsafe, misinformation, loss

Economic

Economic-evaluation results reported as study outcomes.

Examples:

- cost per farmer served or per additional practice adopter
- incremental cost-effectiveness ratio
- benefit-cost ratio or return on investment
- programme cost savings or budget impact

Keywords / cues: cost-effectiveness, benefit-cost, return on investment, cost savings, budget impact

Record farmer income or profit under Livelihood / Food Security Outcome unless the measure is explicitly an economic-evaluation ratio or estimate.

Administrative / System

Changes in agricultural-programme or agricultural-service operations.

Examples:

- efficiency of allocating limited extension visits
- agricultural-insurance claims processing time
- fraud detected
- input stock-out rates or supply-chain performance

Keywords / cues: allocation, targeting, prioritisation, claims, fraud, supply chain, operations

Other

Use only when none of the above categories fit. Explain clearly in the outcome name.

4 Outcome Name

Definition: Specific variable or result being measured. The name should be more precise than the outcome domain.

Examples:

- Sensitivity for rice blast detection
- RMSE of pre-harvest rice-yield forecasts
- Harvested rice yield per hectare
- Irrigation water applied per hectare
- Extension-worker productivity
- Reduction in advisory-programme drop-off
- AI advisory-service enrolment rate
- Weekly chatbot queries per farmer
- Unusable crop-image rate
- Net farm income per season

Keywords / cues: dependent variable, primary outcome, secondary outcome, endpoint, measure

5 Outcome Measure

Definition: How the outcome was operationalized or measured.

Examples:

- AI output vs laboratory-confirmed plant disease or expert field assessment
- predicted yield vs measured crop-cut harvest yield
- completed extension consultations per worker per day from programme records
- share of voice messages listened to beyond a set duration, from call logs
- number of queries per user from system logs
- irrigation volume from metered records
- self-reported income or satisfaction scale

Be concise. Distinguish observed, self-reported, and modeled measures.

Keywords / cues: measured using, reference standard, constructed from, call logs, farm records, crop cuts, survey scale, administrative records

6 Population / Sample for Outcome

Definition: The group or production units for whom this particular outcome was measured. This can differ from the overall study sample.

Examples:

- all enrolled farmers
- farms with harvest measurements
- crop images of sufficient quality only
- plants or animals with reference-standard results
- randomized farmer households
- extension workers who used the chatbot
- women farmers or farms below a reported size threshold

Record the outcome-specific sample size and unit when available. Do not substitute the number of images or observations for the number of farms or farmers.

Keywords / cues: analysis sample, usable images, reference standard available, respondents, subgroup, users, plots

7 Comparison

Definition: What the reported result is compared against.

Possible answers:

- Treatment vs control (randomized)
- AI-assisted agricultural advice vs standard extension
- AI vs human assessor (e.g., agronomist or veterinarian)
- AI-alone vs AI-assisted human
- AI vs reference standard (accuracy; no comparison arm)
- AI threshold A vs threshold B
- Before vs after deployment
- Subgroup A vs subgroup B
- No comparison / descriptive outcome

Example: Villages randomized to AI-assisted crop advice plus regular extension vs villages receiving regular extension alone.

Keywords / cues: compared with, relative to, control, standard practice, reference standard, baseline, difference, treatment effect

8 Result Value

Definition: Main numerical estimate reported for the outcome. Record the number only where practical; do not include units in this field.

Examples: 91.4; 0.78; 450; -12.

If several estimates are reported, use the principal specification (e.g., prespecified primary analysis or prespecified threshold) and explain alternatives in Result Notes. Do not select an estimate merely because it is the largest or most favourable.

Keywords / cues: estimate, effect, sensitivity, specificity, AUC, rate, ratio, mean difference

9 Result Unit

Definition: Unit corresponding to the result value.

Possible answers:

- percent, percentage points, or proportion
- AUC, kappa, or SD
- kg/ha or tonnes/ha
- liters/ha or cubic meters/ha
- kg fertilizer/ha or kg active ingredient/ha
- queries/user/week
- farms/worker/day or hours/ha
- minutes or days
- number of cases, farms, or farmers
- risk ratio, odds ratio, or hazard ratio
- currency or currency/farm/season
- currency per farmer served or per unit of resource saved
- other

Do not confuse percentage points with percent. Retain the source's currency, production basis, and time period. Do not silently convert fertilizer product mass into nutrient mass or pesticide product quantity into active ingredient.

10 Result Interpretation

Definition: Short plain-language description of the quantitative result.

Illustrative example only: Farmers receiving AI-assisted advice harvested 4.5 tonnes per hectare, compared with 4.0 tonnes per hectare in the comparison group.

Keep this factual and concise. Do not add causal interpretation beyond what the study design and authors support. Identify whether an estimate is a group level, change, difference, or ratio.

11 Standard Error / Confidence Interval

Definition: Uncertainty measure reported with the estimate.

Report standard error, confidence interval, or both if available.

Examples:

- SE = 0.07
- 95% CI: 87.1 to 95.0

If not reported: NR.

Keywords / cues: standard error, SE, confidence interval, CI

12 P-value / Statistical Significance

Definition: Reported p-value or significance information. Prefer exact p-values where available.

Examples:

- p = 0.03
- p < 0.001
- statistically significant at 5%
- not statistically significant
- not applicable (no hypothesis test for a descriptive result)

If not reported: NR.

If significance is shown only through stars, report the implied threshold if clearly defined by the paper. Missing significance information is not evidence of a statistically insignificant result.

Keywords / cues: p-value, significance, statistical significance symbols

13 Measurement Timing

Definition: When the outcome was measured relative to the intervention.

Possible answers:

- At the point of field assessment
- End of a 7-week advisory period
- During the trial period
- End of a crop, monsoon, or production season
- At harvest
- 6-month follow-up
- 1-year follow-up

Record the calendar year, season, crop cycle, and forecast horizon where relevant and reported.

Keywords / cues: endline, follow-up, study period, post-intervention, harvest, season, months later

14 Subgroup

Definition: Whether the reported outcome applies to a specific subgroup rather than the full sample.

Possible answers:

- Full sample
- Women or men
- Smallholder farmers, using the source's definition
- Tenant farmers
- Rainfed or irrigated farms
- Low-income or low-literacy beneficiaries
- Remote areas
- High-risk farms
- Usable images only
- Other

If full sample: Full sample.

Keywords / cues: heterogeneous effects, subgroup, by age, by sex, by farm size, by risk group, stratified

15 Actual vs Intended Usage

Use this field only for Adoption or Usage outcomes.

Definition: Clarifies whether the result refers to planned access, initial adoption, actual use, or sustained use.

Possible answers:

- Offered access
- Consented / activated
- Used at least once
- Repeat use
- Average usage
- Sustained usage
- Compliance with protocol

Illustrative example only: Used at least once: 300 of 500 enrolled farmers submitted an agricultural query during the study period.

This field is useful because “offered access” is not the same as “actually used.” Leave it not applicable for other outcome domains.

16 Result Notes

Definition: Short clarification needed to interpret the outcome correctly.

Use for:

- AI threshold or operating point used
- reference standard

- whether unusable images, failed AI outputs, or farms with missing harvest data were excluded from the denominator
- prospective vs retrospective data
- alternative specifications or thresholds
- important denominator definitions
- multiple related estimates
- whether the result is ITT or per-protocol
- whether the result comes from logs, measured field data, programme records, or self-reports
- observed vs modeled outcomes and relevant assumptions
- crop, species, season, weather conditions, or forecast horizon
- any ambiguity

Example: Disease sensitivity computed against expert field assessment; unusable images excluded; estimates for the operational threshold reported separately from alternative thresholds.

Retain a page, table, figure, or section locator for checking the result. Do not use this field for general paper commentary.

Special coding rules

Adoption vs Usage

Do not combine them automatically. Adoption answers: Did the intended user begin using the tool? Usage answers: How much or how often did the user actually use it?

Illustrative examples:

- 60% of eligible farmers activated an AI advisory account → Adoption
- Average of 4 queries per active farmer per week → Usage

Distinguish adoption of the AI tool from adoption of the farming practice it recommends. Practice adoption belongs under Farmer Behaviour / Programme Engagement.

Access vs Usage

Do not treat being offered access as actual usage.

Illustrative examples:

- 1,000 farmers offered chatbot access → Access
- 600 farmers sent at least one query → Adoption
- Average 3 queries per week → Usage

Registration or activation alone does not establish repeated use; preserve the exact participation stage reported.

Accuracy vs Agricultural Outcomes

High diagnostic or predictive accuracy is not evidence of improved production, income, or environmental outcomes. Record accuracy as Diagnostic / Predictive Accuracy.

If the study also measures problem detection, action following advice, production, livelihoods, or environmental outcomes, create separate outcome rows in the corresponding domains. Predicted yield is not observed harvest yield.

Sensitivity and specificity

Record sensitivity and specificity (and AUC, PPV, NPV, precision, recall, or F1) as separate rows for the same target condition and threshold.

If several thresholds are reported, use the prespecified, manufacturer-recommended, or programme threshold as the principal estimate and describe others in Result Notes. Record continuous prediction metrics such as RMSE and MAE separately, retaining their original units and forecast horizon.

Image quality and technical failures

Record the share of images or cases the AI could not process as Technical Performance / Data Quality.

In Result Notes for every accuracy row, state whether these cases were excluded from the denominator. Do not omit sensor failures or unanswered queries merely because successful outputs performed well.

Programme engagement vs Agricultural Outcomes

If an AI system improves engagement with an agricultural programme, record it under Farmer Behaviour / Programme Engagement. Do not treat listening to messages, following advice, or staying enrolled as equivalent to improved yield, income, or environmental conditions. Create separate downstream outcome rows only if those outcomes were measured.

Prediction or prioritisation accuracy

Record prediction quality as Diagnostic / Predictive Accuracy. Record the measured performance of an operational allocation process under Administrative / System where appropriate.

If the study also measures the effect of acting on the predictions, create a separate outcome row. Predicted drought risk is not evidence of improved resilience.

AI alone vs AI assisted human

If a study reports both AI-alone performance and the performance of agricultural workers or farmers using AI, record them as separate rows with different Comparison entries.

Multiple conditions

If a study separately reports results for different diseases, pests, crops, species, production systems, or outcome definitions, create separate rows.

Examples:

- Sensitivity for rice blast

- Sensitivity for bacterial leaf blight

Do not combine different crop yields or disease definitions into a single estimate unless the source explicitly reports a pooled measure.

Multiple time periods

If the same outcome is measured immediately and at longer follow-up, create separate rows when the distinction is important.

Examples:

- AI advisory use during the first month
- AI advisory use during the following crop season
- Harvested yield in the first and second production seasons

Do not interpret changes across seasons as causal effects without accounting for the study's comparison and design.

Data source of the outcome

Record in Result Notes whether each outcome comes from system logs, farm or programme records, independent field measurement, expert assessment, or self-report. Note developer-reported figures without an independent evaluation.

Clearly label modeled or simulated estimates, including modeled environmental benefits, and do not present them as observed field outcomes.

Production resource use and livelihoods

Distinguish harvested yield per unit area from total production and from predicted yield. Record the area and season basis.

Distinguish water withdrawal, water consumption, and water-use efficiency. Reduced pesticide or fertilizer use alone does not demonstrate lower pollution or improved biodiversity.

Distinguish revenue, gross margin, net profit, and household income, including which costs were deducted. Keep currencies and price years explicit when reported.

Intervention packages and missing outcomes

If AI is delivered with training, input subsidies, free devices, or extra extension visits, interpret the reported effect as the effect of that package unless the comparison isolates AI.

Include null, negative, and adverse outcomes. Do not infer zero harms, improved equity, or improved food security when these were not measured.

Output format

Return one table with one row per distinct outcome.

- Tool ID
- Intervention ID
- Source ID(s)
- Short Citation

- AI Tool / Program
- Outcome Domain
- Outcome Name
- Outcome Measure
- Population / Sample for Outcome
- Comparison
- Result Value
- Result Unit
- Result Interpretation
- Standard Error / Confidence Interval
- P-value / Statistical Significance
- Measurement Timing
- Subgroup
- Actual vs Intended Usage
- Result Notes

Do not merge several substantively different outcomes into one row. Use “Not reported” for unavailable information, “Not verified” for claims that could not be checked, and “NR” for unreported statistical uncertainty or significance as specified above. Numerical examples in this rule are illustrative and must not be entered as study findings.